

**MARK TWAIN NATIONAL FOREST
GAOA – COUNCIL BLUFF PRIMARY ELECTRIC
AND WELL IMPROVEMENTS**



PROJECT SPECIFICATIONS

USDA – FOREST SERVICE
401 FAIRGROUNDS ROAD
ROLLA, MO 65401

06/2026

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SECTION 011000 - SUMMARY

PART 1 - GENERAL

1.1 PROJECT INFORMATION

A. Project Identification: Council Bluff Primary Electric and Well Improvements

1. Project Location: This project is located in Iron County Missouri in the Mark Twain National Forest, Potosi-Fredericktown Ranger District recreation site Council Bluff – Wild Boar Ridge Campground.
2. Wild Boar Ridge Campground:
Lat: 37.724361°
Long: -90.940618°

B. Owner: Government or Owners refers to the USDA Forest Service

1. Owner's Representative: The Federal Contracting Officer (CO) and his/her designated representative(S), as named in writing. Contractive Officer (CO) refers to the person assigned by the USDA Forest Service to administer the contract. Soon after Contract award, the Contracting Officer will designate a Contracting Officer's Representative (COR).

C. The Work consists of coordinating with Ameren to extend their existing primary electric line from an overhead power line located near Missouri State Highway DD to the Wild Boar Campground. This includes coming off an existing electric pole on the east side of the highway and installing buried line along the edge of the cleared highway right-of-way to the existing well within the Council Bluff Recreation Area. The corridor will parallel the east side of Missouri State Highway DD for approximately 9,600 feet and then use an "old road corridor" along the top of Wild Boar Ridge leading to the well for approximately 3,100 feet.

The Contractor shall clear the old road corridor along the top of Wild Boar Ridge leading to the well as necessary to allow construction vehicle access. The existing old road corridor contains downed trees and minor brush. **Any tree removal shall occur during the Indiana bat and Northern long-eared bat's hibernation season (between November 1 and March 31).**

A pull box will be placed at approximately every 500-ft along the buried cable. **The electric line along Missouri State Highway DD shall be installed utilizing directional drilling methods. The electric line along the old road corridor can be installed utilizing either directional drilling or trenching.**

The Contractor shall coordinate with the Public Utility (PU) provider, Ameren, to obtain all permits, pay all fees, and provide all labor, materials and equipment to extend the primary distribution system which includes transformers, meter bases, metering, and service connection requirements. This includes all trenching, backfilling, boring, landscaping and other requirements for replacing and extending the primary distribution system necessary to meet project

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requirements. Primary electric installation shall follow Ameren's specifications (latest version) which can be found on their website (<https://www.ameren.com/property/construction>).

Ameren Contact Information:

Michael Swain

mswain@ameren.com

573-438-8023

The work also includes replacing the existing water supply well's pump, drop pipe, pitless adaptor, well cap, and other related appurtenances; connecting the existing well into the new electric system; and other work indicated in the Contract Documents.

- D. The Contractor's electrician shall provide plans for transformer sizing based on the following future needs:

Proposed future electrical needs include:

1. New well house building with a 200 AMP panel
2. 2 host campsites with 50/30/20 AMP RV pedestals
3. 40 campsites with 50/30/20 AMP RV pedestals
4. Septic system with 2 – 230V sewage pumps
5. Existing well with submersible pump

- E. Work is preferred to occur after Independence Day (July 4th) weekend.

1.2 CONTRACTOR'S USE OF SITE AND PREMISES

- A. The government will conduct a preconstruction survey with the Contractor to review and document the existing project conditions prior to construction.
- B. Limits on Use of Site: Limit use of Project site to areas within the Contract limits indicated. Do not disturb portions of the Project site beyond areas in which Work is indicated.
1. The Contractor is responsible for the security of the Contractor's equipment and materials.
 2. Schedule deliveries to minimize on-site storage of materials and equipment.
 3. Existing materials and equipment that are removed as part of the construction operations, and that are not reused or salvaged as Government property, shall become the property of the Contractor and shall be promptly removed from the site.
 4. Utilities are available near the site for construction purposes. The Chapel Hill Beach area has electric, water, and sewer utilities. This is located approximately 3 miles from the Wild Boar Ridge Campground. The water service at Chapel Hill Beach is closed in September and reopened in April.
 5. The site will be open to the public for use during construction. Care will be exercised to adhere to the schedule and to secure the Government facility after hours. The Contractor will be responsible for signing, limiting access, and safety of public and/or government employees in the area.
 6. Keep driveways, loading areas, and entrances serving premises clear and available to Owner, Owner's employees, and emergency vehicles at all times. Do not use these areas for parking or storage of materials.

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C. Staging and storage areas:

1. Areas for staging operations and storage of materials shall be approved by the COR. The Contractor must request in writing for approval from the COR to stage trailers on site.

1.3 WORK RESTRICTIONS

A. Comply with restrictions on construction operations.

1. Comply with limitations on use of public streets, work on public streets, rights of way, and other requirements of authorities having jurisdiction.

B. On-Site Work Hours: Limit work in all project areas to normal business working hours of 8 a.m. to 5 p.m., Monday through Friday, unless otherwise indicated or approved by the COR.

1. No work is permitted without written authorization from the CO on Government Holidays or weekends.

C. Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by Owner or others unless permitted under the following conditions and then only after providing temporary utility services according to requirements indicated:

1. Notify Owner not less than two days in advance of proposed utility interruptions.
2. Obtain Owner's written permission before proceeding with utility interruptions.

D. Noise, Vibration, Dust, and Odors: Coordinate operations that may result in high levels of noise and vibration, dust, odors, or other disruption to Owner occupancy with Owner.

1. Notify Owner not less than two days in advance of proposed disruptive operations.
2. Obtain Owner's written permission before proceeding with disruptive operations.

E. Smoking and Controlled Substance Restrictions: Use of tobacco products, alcoholic beverages, and other controlled substances on Project site is not permitted. Use of any type of marijuana on federal property is illegal.

F. Site Cleanup: Before leaving the Site each day the Contractor shall secure equipment, tools, and materials; pickup and dispose of paper and other debris/construction waste. The Site shall be left in a safe and secure manner with the public protected from construction hazards.

G. Before leaving the Site on Fridays a more complete policing, pickup and securing of the Site shall be performed. The Site shall be left in a neat and orderly manner as determined by the COR with the public protected from construction related hazards.

1.4 INSPECTION OF THE WORKSITE

A. The contractor acknowledges they have taken the necessary steps to ascertain the nature and location of the work and have investigated and satisfied themselves as to the general and local conditions that can affect the work or its cost. Any failure of the contractor to take the actions described and acknowledged in this paragraph will not relieve the Contractor from the responsibility of estimating properly the difficulty and cost of successfully performing the work,

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or for proceeding to successfully perform the work without additional expenses to the government.

1.5 SAFETY

- A. Contractor shall comply with all State and Federal safety regulations during the performance of the work.

1.6 FIELD VERIFICATION

- A. Contractor shall field verify all new and existing dimensions affecting the work of this contract before ordering products.

1.7 DISPOSAL OF MATERIAL

- A. Contractor shall remove and legally dispose of all debris offsite.

1.8 SPECIFICATION AND DRAWING CONVENTIONS

- A. Division 1 – General Requirements: General provisions of the Contract, including General and Supplementary Conditions, apply to all Sections of the Specifications.
- B. Division 2 – Supplemental Specifications: Requirements in Division 02 apply to the Work of all Sections in the Specifications.

1.9 CONTRACT TIME

- A. 120 Calendar Days

1.10 NOTICE TO PROCEED

- A. Performance of the contract shall begin within ten (10) calendar days after receipt of the Notice to Proceed. The Notice to Proceed will be issued by the Contracting Officer (CO) at the pre-work conference.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 011000

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SECTION 012700 – MEASUREMENT AND PAYMENT

PART 1 - GENERAL

1.1 SUMMARY

A. This section includes:

1. Measurement and payment criteria applicable to the work performed.
2. Administrative and procedural requirements necessary to prepare and process Applications for Payment.
3. Defect assessment and non-payment for rejected work.
4. General description of work included under each pay item.

B. Related Requirements:

1. Schedule of Items Form (SOI): The Contractor is required to complete the SOI Form, provided by the Government, as part of their Proposal or Bid.
2. Invoice Processing Platform (IPP) System: All invoices are to be submitted via the electronic IPP System. The Contractor must follow the instructions on how to register and submit invoices via IPP as prescribed in communications from USDA and Treasury. This is a mandatory requirement initiated by the U.S. Department of Treasury and you can find more information at this website <https://www.ipp.gov/index.htm>. Ensure your company has registered at <https://www.ipp.gov/vendors/enrollmentvendors.htm> to establish your account. Offeror must have an established account prior to commencement of work.
3. Certified Payrolls: Certified payrolls for construction work must be submitted to the CO/COR for review at regular intervals as work progresses. At least one week prior to Contractor requesting payment via IPP, all certified payrolls related to the subject pay request work must have been received by the CO/COR. Incomplete or inaccurate payroll submittal will result in rejection of the IPP invoice. Contractor and all subcontractors, through the contractor, shall furnish all certified payrolls covering payments to laborers and mechanics for on-site work. Payroll hourly rates and fringe benefits must be in accordance with the established Wage Determination for the project.

1.2 AUTHORITY

- A. The Contracting Officer's Representative (COR) will take all measurements and compute quantities accordingly.

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1.3 MEASUREMMENT OF QUANTITIES

- A. Lump Sum Quantities (LSQ): These quantities denote one complete unit of work as required by or described in the Contract, including all necessary materials, equipment, and labor to complete the job. They will not be measured.

1.4 PAYMENT

- A. Pay Request Documentation from Contractor: Contractor must submit the following documentation to the CO/COR for review and approval at least seven (7) days prior to requesting a payment via IPP.
 - 1. All Certified Payroll forms from the general Contractor (including payroll from all subcontractors) for work covered by the pay request.
 - 2. A completed draft Schedule of Items (SOI) Form for work covered by the pay request, indicating the percent complete for each line item in the SOI.
 - 3. Written Quality Control testing results if applicable to items in the payment request.
- B. Payment Includes: Full compensation for all required labor, products, tools, equipment, plant, transportation, services, and incidentals; erection, application, or installation of an item of the Work; overhead and profit.
 - 1. "Incidentals" are to be considered as an additional obligation to the other work required for the item(s). Incidental work is not payable directly but is to be considered included in the contract price for the item(s) of work specified.
 - 2. All work not specifically identified as a separate bid item is "Incidental".
- C. The accepted quantities will be paid for at the contract unit price for each item as shown in the Schedule of Items.
- D. Payment will not be made for any of the following:
 - 1. Products or construction placed beyond the lines and levels of the required work.
 - 2. Products or construction determined as unacceptable before or after placement.
 - 3. Products wasted or disposed of in a manner that is not acceptable.
 - 4. Products remaining on hand after completion of the Work.
 - 5. Loading, hauling, and disposing of rejected products.

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PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 012700

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SECTION 012800 - MOBILIZATION

PART 1 - GENERAL

1.1 SUMMARY

- A. This item is intended to compensate the Contractor for operations including, but not limited to, those necessary for the movement of personnel, equipment, supplies, and incidentals to the project site; for payment premiums for bonds and insurance for the project, any required permits to complete the work, and for any other work and operations which must be performed or costs that must be incurred incident to the initiation of meaningful work at the site and for which payment is not otherwise provided for under the contract.

1.2 MEASUREMENT AND PAYMENT

- A. Method of Measurement will be by Lump Sum Quantity (LSQ).
- B. The maximum Mobilization bid price shall not exceed fifteen percent (15%) of the total base bid contract price or optional bid contract price (not including Mobilization).
- C. Progress Payments for Mobilization will be in accordance with the following schedule:
 - 1. The cost of premiums paid for performance and payment bonds are considered incidental to Mobilization and will not be paid for separately. Payment for bond premiums will be in accordance with FAR Clause 52.232-5, Payments under Fixed-Price Construction Contracts, after receipt of the evidence of payment.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 012800

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SECTION 013100 – PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 SUMMARY

- A. This section includes certain administrative procedures for managing and coordinating construction operations.

1.2 GENERAL PROJECT COORDINATION

- A. Coordination of Trades: Coordinate construction operations to provide an efficient and orderly installation of each part of the Work.

1.3 SPILL AND EROSION CONTROL

- A. Spill and Erosion Control Plans: At a minimum, the Contractor shall install erosion control devices as indicated on the plans. The Contractor shall install additional erosion control features to prevent sediment from entering wetlands or waters of the United States and to provide for containment of hazardous materials and unplanned spills in accordance with all applicable federal, state, and local laws.

1.4 ADMINISTRATIVE AND SUPERVISORY PERSONNEL

- A. In accordance with FAR 52.236-6, at all times during performance of this contract and until the work is completed and accepted, the Contractor shall directly superintend the work or assign and have on the worksite a competent supervisor who is satisfactory to the Contracting Officer and has the authority to act for the Contractor.
- B. In addition to meeting the requirements of FAR 52.236-6 for an on-site Project Superintendent, the Contractor shall provide other administrative and supervisory personnel as required for the performance of the Work.

1.5 CONFERENCES AND MEETINGS

- A. Pre-construction Conference: The Contracting Officer or Contracting Officer's Representative will schedule a pre-construction conference before starting construction at a time and place agreeable and convenient to both the Contractor and the Government and will review responsibilities and personnel assignments in accordance with FAR 52.236-26.

1.6 SUBMITTALS

- A. Staff Names: Within 5 calendar days after Notice to Proceed, submit a list of principal staff assignments including the Superintendent and other primary personnel. Identify individuals by

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name, duties and responsibilities, and business/cell telephone numbers. Post copies of this list in the temporary field office (if present).

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 GENERAL COORDINATION PROVISIONS

- A. Inspection of Conditions: Prior to installations, require the installer of each major component to inspect both the substrate and conditions under which work is to be performed.
- B. Construction in Progress: Keep construction in progress, adjoining materials in place, and clean during handling and installation. Apply protective enclosures or coverings for protection from damage or deterioration.
- C. Completed Construction: Clean completed construction and provide maintenance to prevent damage, soiling, or other deterioration through the remainder of the construction period. Adjust and lubricate operable components to assure operability without damage.
- D. Limiting Exposures: Supervise construction operations to prevent exposure of any part of construction, completed or in progress, to harmful, dangerous, damaging, or otherwise deleterious conditions during the construction period.

END OF SECTION 013100

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SECTION 013250 – CONSTRUCTION PROGRESS SCHEDULE

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Preliminary Schedule

1.2 SUBMITTALS

- A. Submit a comprehensive, fully developed schedule within 10 days of Notice to Proceed date.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 CONTENT

- A. Preparation: Indicate each significant construction activity and the location on the site separately. Identify first workday of each week.
- B. Show complete sequence of construction by activity, with dates for beginning and completion of each element of construction.
- C. Identify each item by specification section number.
- D. Show accumulated percentage of completion of each item, and total percentage of Work completed, as of the first day of each week.
- E. Contractor to revise schedule if Work is more than one week behind schedule.

END OF SECTION 013250

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SECTION 013300 – SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section and FAR 52.236-21 include requirements for Shop Drawings, as defined by FAR 52.236-21 and Designs. Shop drawings and Designs may include drawings, specifications, diagrams, layouts, schematics, descriptive literature, illustrations, schedules, performance, and test data furnished by the Contractor to explain in detail specific portions of the work required by the Contract.
- B. This section includes requirements for Samples. Samples may include mailed samples, field samples or mock-ups of full-size physical examples erected on-site or elsewhere.
- C. This section includes requirements for Product Data. Product Data includes manufacturer's standard catalogs, manuals, pamphlets and other printed materials, and includes but is not limited to the following: product specifications and installation instructions, color charts and catalog cuts, rough-in diagrams and templates, wiring diagrams, performance curves, operational range diagrams, mill reports, test reports, shop drawings and code approvals.
- D. This section includes other quality assurance submittals such as certifications of compliance or conformance, manufacturer's instructions and field reports, and mix designs.

1.2 GENERAL SUBMITTAL REQUIREMENTS

- A. Coordination: Coordinate preparation and processing of submittals with performance of construction activities and with the schedule.
 - 1. Allow 15 calendar days each for submittal review, corrections following the initial review and re-submittal review before activities scheduled after the submittal approval.
 - 2. Incomplete submittals will be returned to the Contractor with no further review. Delays due to incomplete or rejected submittals will not be excused and no compensation in the form of monetary or time will be provided to the Contractor.
 - 3. When the Contractor desires to submit for approval a product for an “approved equal”, the Contractor **MUST** provide documentation of the original specified product and the Contractor proposed product in a side-by-side comparison, such as a table, which clearly demonstrates the Contractor proposed product meets or exceeds the originally specified product. If this product comparison is not done the first time it is submitted, the proposed product will be rejected, **and the Contractor will be required to provide the specified product. The Government is not obligated to review a re-submittal for a rejected product.**

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4. Submittal Transmittal: Transmit each submittal from the Contractor to the Government by use of one transmittal form per submittal. Transmittal format to be approved by COR and shall include the following information on the transmittal form.
 - a. Project name and Forest Service contract number.
 - b. Date.
 - c. Name, address and telephone number of firm or entity that prepared the submittal.
 - d. Names of subcontractor, manufacturer, supplier, etc. as applicable.
 - e. Category and description of submittal.
 - f. Section of Specifications or Drawing sheet number the Submittal is representing.
 - g. Certification by Contractor stating that submittal complies with the Contract, or statement of deviations from the requirements of the Contract.
 - h. Signature of transmitter.
 - i. Space to record Contractor's review and approval markings, and for Government response.
- B. Submit electronic copies of each submittal to the COR for review unless otherwise stated.

1.3 SHOP DRAWINGS AND DESIGNS

- A. Submit originally prepared information, drawn accurately to scale.
- B. Submittals: Submit three blue-line or black-line prints of each drawing submittal. One print will be marked with action taken and returned. Distribution: When submittal is approved, the Contractor shall prepare and submit three final copies.
- C. Shop drawings shall be done in AutoCAD format. Upon final approval the drawings shall be sent on a thumb drive or via email to the COR.
- D. Specifications shall be done in Microsoft Word format and a copy sent on a thumb drive or via email to the COR.

1.4 PRODUCT DATA

- A. Collect Product Data into a single submittal for each system or element of construction. Mark each copy to show specific product choices and options applicable to the project. Product Data shall include the following information, where applicable:
 1. Manufacturer's printed recommendations.
 2. Compliance with recognized trade association standards.

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3. Compliance with recognized testing standards.
 4. Applicability of testing agency labels and seals.
 5. Notation of dimensions verified for fit by field measurements.
 6. Notation of coordination requirements.
- B. Preliminary Submittal: Prior to submittal of complete Product Data, submit a preliminary single copy of that part of Product Data when selection of options is required, such as for color charts. Preliminary submittal will be returned, with selection noted, for the Contractor's use in subsequent submittals.
- C. Submittals: Unless otherwise indicated, submit 3 copies of each Product Data submittal. Two copies will be retained, and the remaining copy will be marked with action taken and returned. The COR may allow electronic copies to be transmitted in lieu of hard copies.
- D. Material and Product substitution shall be in accordance with FAR 52.211-6 and Section 013300-1.2. A.3.

1.5 SAMPLES

- A. Submit full-size, fully fabricated samples, cured and finished in the manner specified. Samples shall be physically identical to the material or product proposed for use.
- B. Preliminary Submittal: Where color, pattern, texture or similar characteristics are to be selected from a manufacturer's range of standard choices, submit a preliminary single set sample of available choices when required by the CO.

1.6 OTHER QUALITY ASSURANCE SUBMITTALS

- A. Submit other quality assurance submittals in compliance with requirements in the individual specification sections.

1.7 REVIEW ACTION ON SUBMITTALS

- A. Except for submittals for the record or for information or for another purpose where no action and return is required, the Government will review submittals and mark returned copies to indicate action taken within a maximum time period of 15 calendar days.
- B. Submittals that do not contain an appropriate marking of approval by the Government shall not be used for construction.

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PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 013300

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SECTION 014000 – QUALITY CONTROL

PART 1 - GENERAL

1.1 SUMMARY

- A. This section, FAR 52.246-1 Contractor Inspection Requirements and FAR 52.246-12 Inspection of Construction, include requirements for Contractor quality control services used to verify quality assurance requirements specified elsewhere in the Contract.
- B. The Government reserves the right to conduct independent tests.
- C. This section includes requirements for quality control services such as piping pressure tests, water sampling, concrete mix and strength testing, and bedding and sand gradation tests (this is not an all-inclusive list of tests that may be required) per State of Missouri Agency Requirements – MO DOT and MO DNR– as they apply to the work being performed.
- D. The quality control services include tests and related actions, including reports, performed by the Contractor, independent testing agencies or governing authorities.

1.2 QUALIFICATIONS OF INDEPENDENT TESTING AGENCIES

- A. A qualified independent testing Agency shall be an accredited entity engaged to perform tests or inspections, and to report on and, if required, to interpret results of those tests or inspections.

1.3 SUBMITTALS

- A. **REPORTS:** The Contractor shall submit a certified written report, in duplicate (or electronic format), of each test, inspection or other quality control service to the Government.
- B. Written reports shall include but not be limited to the following:
 - 1. Date of issue.
 - 2. Project title and number.
 - 3. Name, address, and telephone number of testing agency.
 - 4. Dates and locations of samples and tests or inspections.
 - 5. Names of individuals making the test or inspection.
 - 6. Designation of the work and test method.
 - 7. Identifications of product and specification section.
 - 8. Complete test or inspection data.

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9. Test results and an interpretation of test results.
10. Ambient conditions at the time of sample taking and testing.
11. Comments or professional opinion on whether tested or inspected Work complies with Contract Document requirements.
12. Name and signature of laboratory inspector.
13. Recommendations on retesting.

1.4 MEASUREMENT AND PAYMENT

- A. Method of Measurement will be by Lump Sum Quantity (LSQ).
- B. Payment will be full compensation for work prescribed in this Section. Payment of lump sum pay items will be prorated based on the progress of the work under this Section.
- C. Progress Payments and Final Payment for Quality Control will be in accordance with FAR 52.232-5 Payments under Fixed-Price Construction Contracts. The Government shall make progress payments monthly as the work proceeds on estimates of work accomplished which meets the standards of quality established under the contract, as approved by the Contracting Officer.

1.5 RESPONSIBILITIES

- A. Contractor Responsibilities: The Contractor shall provide for tests, inspections and other quality control services (such as well water, electrical, etc. tests and inspections) where specified in the Contract or when required by authorities having jurisdiction such as the State of Missouri Agencies—MODOT and MODNR. Costs for these services are included in the Contract price. The Contractor shall maintain complete inspection records and make them available to the Government.
- B. The Contractor shall employ and pay for a qualified independent testing agency (Agency) to perform the quality control services.
- C. Associated Services: The Contractor shall cooperate with others performing required tests, inspections and other quality control services, shall provide access to the work, and shall furnish incidental labor and facilities necessary to facilitate inspections and tests.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION 014000

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SECTION 017700 – CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for Phased Project completion and Final Contract closeout, including, but not limited to, the following: The Government reserves the right to conduct independent tests.
 - 1. Systems Commissioning including Demonstration and Training.
 - 2. Record Documents including Record Drawings and Product Data.
 - 3. Substantial and Final completion procedures.
 - 4. Final cleaning.

1.2 DEFINITIONS

- A. Substantial Completion - The point at which the project, or a major phase of the project, is complete such that it can be safely and effectively used by the public without further delays, disruption, or other impediments.
- B. Final Completion – The point at which the project is 100% complete, accepted by the Government and ready for final payment.

1.3 RECORD DOCUMENT SUBMITTALS

- A. Record Drawings (a.k.a. as-built drawings):
 - 1. Maintain a clean, undamaged set of blue or black line prints of Contract Drawings and Shop Drawings. Mark the set to show the actual installation where the installation varies from the Work as originally shown. Mark all drawings that are affected by the variation(s) fully and accurately. Where Shop Drawings are used, record a cross-reference at the corresponding location on the Contract Drawings. Give particular attention to concealed and buried elements that would be difficult to measure and record at a later date.
- B. Record Product Data: Maintain one copy of each Product Data submittal. Note related Change Orders and markup of Record Drawings and Specifications.
 - 1. Mark these documents to show significant variations in actual Work performed in comparison with information submitted. Include variations in products delivered to the site and from the manufacturer's installation instructions and recommendations.
 - 2. Give particular attention to concealed products and portions of the Work that cannot otherwise be readily determined later by direct observation.

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1.4 FINAL ACCEPTANCE PROCEDURES

- A. Submittals Prior to Final Completion Inspection and Final Acceptance: Submit the following a minimum of 10 days before scheduling a final inspection for determining Final Completion and Acceptance:
1. Submit a final Schedule of Items (SOI) form to reflect the final payment values in accordance with Section 012700 "Measurement and Payment."
 2. Submit all remaining Certified Payrolls for the project not previously submitted.
 3. Submit completed, signed Contract Release form.
 4. Submit a signed copy of the Substantial Completion inspection list of items to be completed or corrected (punch list), endorsed and dated by Government CO/COR. Certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance.
 5. Submit all remaining specific warranties, workmanship bonds, maintenance agreements, final certifications and similar documents not previously submitted.
 6. Submit all remaining record documents and similar final record information not previously submitted.
 7. Deliver tools, spare parts, extra stock and similar items not previously delivered.
 8. Complete final clean-up requirements including touch-up painting of marred surfaces, adjustments, lubrication, etc.
- B. Final Inspection Procedure: The Government will inspect the Work upon receipt of notice from the Contractor that the Work, including punch list items from earlier inspections, has been completed.
1. Upon completion of inspection, the CO will notify the Contractor of Final Acceptance and Final Payment or will advise the Contractor of Work that it is incomplete or of obligations that have not been fulfilled and are required for Final Acceptance.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 FINAL CLEANING

- A. Perform final cleaning. Conduct cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations. Do not burn waste materials. Do not bury debris or excess materials on Government property. Do not discharge volatile, harmful, or dangerous materials into drainage or septic systems.
1. Complete the following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project:

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- a. Clean Project site of rubbish, waste material, litter, and other foreign substances.
 - b. Clean exposed exterior and interior hard-surfaced finishes to a dirt-free condition, free of stains, films, and similar foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to their original condition.
 - c. Remove labels that are not permanent.
 - d. Leave Project clean and ready for occupancy.
- B. Construction Waste Disposal: Comply with waste-disposal requirements in all applicable specification sections and contract clauses. Remove waste materials from the site and dispose of it lawfully.

END OF SECTION 017700

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SECTION 221114 – WATER SERVICE

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes water-distribution piping and specialties for water services including, but not limited to, water lines, well pumps, and disinfection.

1.2 DEFINITIONS

- A. Water Service: Exterior domestic-water piping.
- B. The following are industry abbreviations for piping materials:
 - 1. PVC: Poly Vinyl Chloride pipe
 - 2. GS or GIP: Galvanized steel or Galvanized Iron Pipe – used interchangeably.
- C. Pump Testing includes well pump testing, pump test reporting, and chlorination.
- D. Missouri Department of Natural Resources Division of Environmental Quality “Standards for Noncommunity Public Water Supplies, 1982” is the same as “Standards”.

1.3 SUBMITTALS

- A. Product Data: For all piping specialties to include, but not limited to, all valves, transition fittings, GIP pipe and fittings and any other item used in the construction of the water distribution system.
- B. Pump Test data written results
- C. Pump operation and maintenance manual and pump curve
- D. Water Sample written results clearly stating MCL for each contaminant, corresponding test result with units and clearly stating if the contaminant exceeds the MCL.
- E. Proof of Laboratory’s current EPA certification for PFAS analysis
- F. Licenses:
 - 1. Provide a copy of current Missouri Well Driller License.
 - 2. Provide a copy of current Missouri Electrician License.

1.4 QUALITY ASSURANCE

- A. Regulatory Requirements:

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1. Comply with standards of authorities having jurisdiction for potable-water-service piping, including materials, installation, testing, and disinfection.
2. Piping materials shall bear label, stamp, or other markings of specified testing agency.
3. Comply with NSF 61 for materials for water-service piping and specialties for domestic water.
4. Comply with lead free requirements for all potable water fixtures as per Section 1417 of the Safe Drinking Water Act.
5. AWWA Compliance: Comply with AWWA A100 for water supply wells.
6. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

PART 2 - PRODUCTS

2.1 WELL PUMP, CAP, PITLESS ADAPTOR, DROP PIPE, CHECK VALVES, TORQUE ARRESTORS

- A. The submersible well pump and controls shall be capable of delivering 15 gallons per minute at 680 feet of total dynamic head. Pump shall be at least a 5HP stainless steel shell and motor, single-phase, 230-volt, with check valve. Pump HP may need to be adjusted after test pumping.
 1. Goulds e-GS High Capacity Series 45GS75 or approved equal.
 2. The existing well pump is assumed to be a Burke Series SN-34C Hydro-Sub 5 HP 230V.
- B. Drop pipe shall be 1.5" diameter, Galvanized steel, Schedule 40, pipe shall be in accordance with ASTM A53 Grade B and have threaded and coupled fittings in accordance with ANSI B16.1. Pipe shall be NSF 61 approved and rated not less than 200 psig.
- C. Provide compatible galvanized reducers as required at connections to pump and pitless adapter.
- D. Check valves shall be of the same size as the drop pipe and placed every 200' with one being just above the pump. Check valve body shall be lead-free bronze or brass, with stainless steel spring material and be NSF-61 approved. Valves must be rated to at least 200 psi water pressure and designed for either horizontal or vertical orientation. Valve body must be rated to handle the weight of the pump and drop pipe assembly, water column and pressure.
- E. Torque Arrestors shall be flexible molded rubber adjustable torque arrestors that adjusts to fit 4" to 8" diameter well casings, placed at each check valve location. Each torque arrestor shall include 2 each stainless-steel hose clamps to secure the torque arrestor to the drop pipe. Torque arrestors are to also act as well-centralizers.
- F. Well cap shall meet State of Missouri requirements but must be at least a metal sanitary cap that fits over the top of the casing forming a waterproof and vermin seal with threaded hole for electrical PVC conduit. Well cap shall have an integral vent screen. The vent shall terminate in a downturned position, at or above the top of the casing or pitless unit in a minimum 1 ½ inch

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diameter opening covered with a 24 mesh, corrosion resistant screen. The pipe connecting the casing to the vent shall be of adequate size to provide rapid venting of the casing. There shall be no other openings in the well cap. The well cap shall be lockable, Baker-Monitor well cap part number 6WTCL or approved equal.

- G. Pitless adapter shall be a weld on type or approved equal, State of Missouri approved, with minimum 3.5' bury depth.

2.2 JOINING MATERIALS

A. Transition Couplings:

- 1. Underground Piping, NPS 2.5" and Smaller: Manufactured fitting or coupling same size as, with pressure rating at least equal to and ends compatible with, piping to be joined.

- B. Pipe thread compound shall be Blue Magic Industrial Grade or approved equal. It shall be NDF 61 approved for potable water, grit free, lead free and non-hardening once applied to the pipe threads.

PART 3 - EXECUTION

3.1 FIELD QUALITY CONTROL

- A. Performance Testing: Conduct final pumping tests after new well pump, drop pipe, and appurtenances have been installed. Test pump the existing well as per Missouri Department of Natural Resources requirements and record on FS provided form (modify as needed for time of pumping/recharge).

- 1. Provide discharge piping to conduct water to locations where disposal will not create a nuisance or endanger adjacent property.
- 2. Measure elevation to water level in well.
- 3. Test Pump: Variable capacity test pump with capacity equal to maximum expected yields at pressure equal to drawdown in wells, plus losses in pump columns and discharge pipes.
- 4. Start and adjust test pumps and equipment to required pumping rates.
- 5. Record readings of water levels in wells and pumping rates as per the FS provided form Well Pump Test and Recovery Data Sheet.
- 6. Record well recovery water levels in well as per the FS provided Well Pump and Recovery Data Sheet until the static water level has become stable.
- 7. Remove sand, stones, and other foreign materials that may become deposited in wells after completing final tests.

- B. Water Analysis Testing: Make bacteriological, physical, and chemical analyses of water from the well and report the results. Samples of water shall be collected in laboratory provided sample bottles and tested by a State of Missouri certified laboratory for the following constituents:

- 1. Antimony
- 2. Arsenic
- 3. Aluminum
- 4. Asbestos
- 5. Barium

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6. Beryllium
7. Cadmium
8. Chloride
9. Chromium
10. Copper
11. Corrosivity
12. Cyanide
13. Fluoride
14. Foaming agents
15. Iron
16. Lead
17. Manganese
18. Mercury
19. Nitrate
20. Nitrite
21. Odor (Threshold Number)
22. Per- and Polyfluoroalkyl Substances (PFAS)
 - a. Must submit proof of Lab's current EPA certification for PFAS analysis
 - b. Analyze sample using a US EPA approved analytical method (EPA Methods 533 or 537.1)
23. pH
24. Selenium
25. Silver
26. Sulfate
27. Thallium
28. Total Coliform (2 tests, at least 24 hours apart)
29. Total Dissolved Solids
30. Turbidity
31. Water Temperature (at the site)
32. Zinc

3.2 CLEANING

- A. Disinfect water supply wells according to AWWA A100 and AWWA C654 before testing well pumps.

3.3 PROTECTION

- A. Water quality protection: Prevent well contamination, including undesirable physical and chemical characteristics.
- B. Provide casings, seals, sterilizing agents, and other materials to eliminate contamination; shut off contaminated water.

3.4 PIPING APPLICATIONS

- A. Do not use flanges, unions, or keyed couplings for underground piping.

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3.5 JOINT CONSTRUCTION

A. Make pipe joints according to the following:

1. Dissimilar Materials Piping Joints: Use adapters compatible with both piping materials, with OD, and with system working pressure.
2. Galvanized Iron Pipe: Use proper thread compound.

END OF SECTION 221114

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SECTION 260010 – GENERAL TECHNICAL PROVISIONS

PART 1 - GENERAL

1.1 DESCRIPTION OF THE PROJECT

- A. Furnish all labor and materials necessary for the installation, connection, testing, and adjustment of all electrical equipment specified herein, shown, or notes on the drawings, specified or required in other portions of this specification and delivered to the Government as a complete installation in all respects ready for use.
- B. Coordinate all sections of Division 26, within the division and with all other divisions. All general technical provisions apply to all sections of Division 26.

1.2 PROJECT SCOPE FOR ELECTRICAL WORK

- A. The contractor shall coordinate with the Public Utility (PU) provider, Ameren, to pay all fees and provide all labor, materials and equipment to extend their primary distribution system which includes transformers, meter bases, metering, and service connection requirements. This includes all trenching, backfilling, boring, landscaping and other requirements for replacing and extending the primary distribution system necessary to meet project requirements.
- B. It is the Contractor's responsibility to perform all coordination to determine Ameren's requirements at the time of bidding and include all costs within the contract amount. The Contractor shall provide the COR with copies of all permit applications and other requirements provided by the utility company for review.
- C. The Contractor shall provide and install the electrical system as shown on the drawings and indicated in the specifications.
- D. The electrical drawings are schematic in nature and not intended to show exact routing, or all equipment, devices, connections, and materials that is required. The Contractor is required to provide all additional equipment, materials, and labor necessary to complete the electrical system and make it fully operational in accordance with the Contract and National Electric Code (NFPA 70) latest edition.
- E. Transformer, meter/panel locations, and feeder corridors shall be staked in the field and reviewed and approved by the COR prior to the start of any electrical work.

1.3 REFERENCES

- A. 013300 – Submittal Procedures
- B. 260050 – Basic Materials and Methods
- C. 260400 – Underground Power Distribution

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1.4 QUALITY ASSURANCE

- A. Codes and Standards: All materials shall conform to the applicable current standards (standard rules, regulations and specifications), Institute of Electrical and Electronic Engineers (IEEE), Insulated Power Cable Engineers Association (IPCEA), NFPA-70 National Electric Code (NEC). All materials shall bear the label of, or be listed by, the Underwriter's Laboratory, Inc., (U.L.). Where conflicts exist between any of the above standards, the standard which is the most stringent shall take precedence.
- B. Errors and Omissions: Any errors found in the plans or specifications shall be brought to the attention of the COR. Contractor is responsible for execution of all work to meet applicable codes. Discovery of errors does not relieve contractor from responsibility to perform all work to meet standards and codes.
- C. Resolve all conflicts with applicable regulations. Prepare and submit to the CO all additional clarifying details including substitute methods and materials as needed to secure the required approval and resolve conflicts.
- D. All work shall be done by electricians licensed by the State of Missouri.

1.5 DRAWINGS

- A. For clarity and legibility, the electrical plan is a broad schematic representation of the overall electrical system. The size and location of equipment is approximate. Make use of all the data in the contract documents and verify information against actual field conditions. Install materials and equipment to conform to the available space. Maintain required clearances and satisfy the requirements of the governing codes.
- B. Electrical drawings and layout may not necessarily suit the requirements of a particular manufacturer's equipment. Modify wiring and location, provide additional materials and work as required for proper installation in accordance with NEC or manufacturer's instructions.
- C. Submit proposed deviations to COR for approval and note change(s) on record drawings prior to making such modification.

1.6 SUBMITTALS

- A. Submit under provisions of Section 013300 – Submittals.

1.7 DELIVERY AND STORAGE

- A. Deliver and store materials in original containers, protected from the elements and theft, and accessible for inspection.
- B. Store items subject to moisture in a dry, heated place.
- C. Protect material against dirt, water, chemical, or mechanical injury.

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1.8 EXCAVATION AND BACKFILL

- A. Typical underground utility electrical details for the secondary power system are shown on the drawings. For the primary distribution system follow utility provider's requirements.

1.9 INSPECTION AND TESTS

- A. At the pre-final inspection, conduct tests to determine that a complete workable electrical system has been installed conforming to the drawings, applicable codes, and specifications.

PART 2 - PRODUCTS

2.1 GENERAL

- A. See Section: 260050 Basic Materials and Methods and 260400 Underground Power Distribution
- B. Provide all products show complete with accessories and mounting materials.

2.2 COMMON MANUFACTURER

- A. Provide all materials of the same type from one manufacturer for all products used on the project. Example: Do not supply power panels from multiple manufacturers.

PART 3 - EXECUTION

3.1 GENERAL

- A. Follow manufacturer's directions in delivery, storage, protection and installation of equipment and materials. Obtain submittal approval prior to installation of materials or equipment.
- B. Notify COR of any conflicts between manufacturer's instructions and contract documents.

3.2 WIRING METHODS

- A. Underground conduit is 3" Schedule 40 PVC or equivalent HDPE unless noted otherwise.
- B. Install equipment and materials, align, level and adjust for satisfactory operation. Install equipment so that all parts are easily accessible for inspection, operation, and maintenance and repair.
- C. Structural Openings: Obtain written permission of the COR before cutting or piercing structural members. Install sleeves to accommodate the penetration. All openings in structure made by parts piercing waterproofing shall be sealed and made watertight.

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- D. Sealing: Seal equipment raceways, boxes, etc., exiting through structures exposed to the weather and make watertight and insect-proof. Protect equipment outlets and conduit openings at all times that work is not in progress; utilize temporary plastic plugs or caps designed for the purpose. All penetrations through fire rated walls and floors shall be fire sealed with UL listed fire rated material.
- E. Required Clearances: Comply with NEC Part 110-26. No grounded equipment will be allowed in this clearance space.
- F. Mounting Heights: Conform to the following mounting height requirements unless otherwise indicated on the drawings by detail or by dimension shown. Mounting heights indicate for pendant mounted devices are to the bottom of the device' mounting heights indicated for outlets and similar devices are to the center of the box unless otherwise indicated.

DEVICE	MOUNTING HEIGHT
Toggle switches and push-button stations	48-inches
Receptable outlets and similar devices	18-inches or ABA height required
Panelboards and control panels	72-inches
Junction box on wall	12 inches or as needed for equipment

- G. Junction Boxes and Pull Boxes: Install junction and pull boxes in locations that will be accessible after completion of the installation.
- H. External Junction Box Sealing: Seal conduits entering external junction and outlet boxes on outer side of structure with duct seal for three inches to prevent moisture entrance.

3.3 RACEWAYS

- A. Raceway Sealing: Seal raceway where entering through structure with duct seal for three inches to prevent moisture entrance.
- B. Sizes: Use raceways no smaller than $\frac{3}{4}$ inch except that $\frac{1}{2}$ inc may be used for switch legs, and ground conductors to ground rods, or as shown on drawings.
- C. Raceway Exposed: Run exposed raceways in straight lines, at right angles, or parallel with building and equipment lines.
- D. Underground Conduit:
1. Install raceways to provide not less than 24 inches to finished grade unless otherwise shown on drawings. Pitch to drain; avoid trapped runs where possible. Grade trenches and place select material to provide uniform trench bottom for raceway support.
 2. See 260400 for detailed underground conduit and conductor installation.
- E. Thoroughly clean the interior of all underground conduits and other raceways of all foreign material by compressed air or swabbing. All exposed conduit shall be clean and in new condition free of dents and machine markings.

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3.4 STUB-UPS

- A. Extend rigid steel conduit stubs at least 1 foot above slab or fill. At motor control panels and similar stub-ups, extend conduit stubs at least 2 inches above base of panel or into panel and install insulated grounding bushing and grounding connection on each conduit. Locate all stub-ups for equipment connection as close as practical to the unit.
- B. Extend rigid steel conduit stubs 6 inches above floor for connection to motors unless otherwise shown on drawings.
- C. Provide duct seal.

3.5 PROTECTION

- A. Cap raceways immediately upon rough-in. Utilize temporary plastic caps designed for the purpose.

3.6 CONDUCTORS

- A. General: Circuit all feeders and branch circuits as shown on the drawings. Provide copper conductors for building wiring.
- B. Conductor and associated conduit sizes are shown on drawings. Where conduit size is not indicated follow NEC requirements for sizing conduit.
- C. The complete circuit shall be size indicated on drawings, unless otherwise noted.
- D. Control and Motor Power Circuit Conductors: Provide stranded copper, THWN conductors for control and motor power circuits in conduit, unless otherwise shown on drawings.
- E. Provide color coding to match control panel terminal block conductors. Control panel wiring diagrams will be furnished to show conductor colors.
- F. Minimum Conductor Size: No. 12 AWG copper minimum size for branch circuits and control circuits unless otherwise specified, shown on the drawings, or required by overcurrent protection.
- G. Underground Conduits and pull vaults: See Section 260400.
- H. 600V Conductors: Provide one of the conductor types indicated for the function and location listed below unless otherwise indicated on the drawings. Provide ground and neutral wires identical in size to circuit wires unless otherwise shown on drawings. At the Contractor's option, conductors with insulation systems rated for high operating temperatures may be substituted for lower temperature rated conductors; however, no reduction in conductor size will be permitted from that indicated on the drawings. When using smaller overall diameter wire than THW, do not reduce conduit size below that required for type THW insulated conductor.

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I. Insulation type:

LOCATION	THHN THWN	XHHW OR THHN	XLP-USE-2
Lighting Circuits, Interior General	X	X	
Inside Lighting Fixture Wiring Compartments		X	
Branch Receptable Circuits	X	X	
Motor Circuits and Controls	X	X	
Power Outlets	X	X	
Underground Feeders in Conduit			X
Underground Conductors in Conduit at RV Pedestal			X

J. Identification:

- Furnish factory color coded 600V wire No. 6 AWG and smaller as follows:

	120/240V
	1-PHASE
Phase A	Black
Phase B	Red
Phase C	
Neutral	White
Ground	Green

- Phase code wire size No. 4 AWG and larger utilizing waterproof, non-metallic, colored pressure sensitive tape at all boxes, switchboards, panel boards and at equipment served.

3.7 IDENTIFICATION NAMEPLATES (SEE 260050 FOR NAMEPLATE CONSTRUCTION)

A. Provide engraved plastic nameplates for all electrical equipment as follows:

- Panelboards. Indicate panel name, voltage, and phase, number of wires.
- Receptacles

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- B. Identify all exposed feeder and branch conduit and control cables with nameplates. Designate the apparatus served and the voltage of the circuit. Identify each conduit once in each room, or space, where it is exposed.
- C. Mount labels with machine screws, except where screw penetration will injure equipment, contact or epoxy cement or ty-raps may be used.
- D. Splices in splice vaults: Indicate circuit number's point of origin, voltage and phase if power cable, and point of disconnecting means, see 260400.
- E. Provide 4" x 6" nameplate at each power panel service or control center with the following designations:
 - 1. Electrical Contractor.
 - 2. Panel name, Installation date and system voltage, phase, number of wires, (120/240 Volt, 1 phase,) example, and amperage of panel bus.

3.8 WIRE NUMBERS

- A. Label each end of each power and control conductor, IT wiring cables, control cables, entering each motor control panel on the load side of the connection terminal block and at each external controlled or pilot device (motor, sensor, float switch, disconnect, etc.) in its connection junction box, and any intermediate terminal or junction box with a machine numbered band.
- B. Equipment or panels that are being replaced shall have wire numbers put on each conductor prior to removing equipment and a drawing of the connections be made for reconnection of the conductors.
- C. Label all branch circuits in power distribution panels and all junction boxes and at all terminations, disconnects, outlets etc., with numbers corresponding to circuit numbers.
- D. Wire numbers shall correspond to control panel schematic and wiring diagram's external connection terminal numbers.

3.9 CONDUCTOR INSTALLATION

- A. Keep all conductors within the manufacturer's allowable tension limits during installation. Use lubricants for wire pulling intended for use with the insulation and raceway material. Observe cable manufacturer's and industry standard cable bending radius recommendations. Do not install conductors at temperatures less than 30 degrees F.

3.10 SPLICING AND TERMINATIONS

- A. General: Provide all splices and terminations including all required hardware for a complete installation.
- B. Provide overall insulation equal to that of the conductors utilized. Insulation may be vinyl plastic tape, preformed insulating covers, cast-in-place encapsulating, insulating resin, or heat shrinkable

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tubing as appropriate for the installation of a completely insulated system. See Splicing requirements in Section 260400.

3.11 PULL VAULTS

- A. Inspect actual field conditions at the proposed location for each splice vault and verify that it is free from interference with other utilities and free of flooding due to the characteristic flow of surface water. Make minor relocations required to clear obstructions and minimize flooding.
- B. See 260400 for pull vaults

3.12 EQUIPMENT SUPPORTS

- A. Raceway Supports: Support raceways with wall brackets, perforated channel, pipe hangers, or ceiling trapezes. Run conduit parallel and perpendicular with each other and building lines. Wire, chain, or perforated plumber's strap are not acceptable for raceway supports. Support conduit 8'-0" on centers max. and within 18" of termination. Use trapeze hangers, pipe hangers, or channel to support two or more conduits per run.
- B. Supports: Provide hangers or other devices such as pads, channels, struts, joists, anchors, etc., necessary for the support of electrical equipment. Provide the design, fabrication and erection of supplementary structural framing required for attachment of hangers or other devices supporting electrical equipment. Show on shop drawing supplementary framing including design loads, member size, and location. Verify that dimensions are suitable for the equipment furnished. Provide additional strength when equipment furnished is heavier than that specified.
- C. Provide unistrut mounting channel for equipment to walls floors. Provide stainless steel material for corrosive or outdoor locations.

3.13 WIRING AND DISCONNECTS

- A. Provide power wiring including service to apparatus and line voltage (or low voltage if specified) control wiring.
- B. Provide branch circuit wiring to each fixture and motor sized as shown on the drawings. Verify motor characteristics prior to installing wiring. After connecting, check for proper rotation.
- C. Contractor shall provide motor disconnects, where specified, shown or required by code, under this division of the specifications.

3.14 GROUNDING SYSTEM

- A. General: Provide all grounding systems and connections above grade by the fusion-welding process or with bolted solderless connectors, in compliance with UL 467, and those below grade shall be made by a fusion-welding process.

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- B. Ground all line and low voltage electrical systems completely and effectively as required by code and as specified herein.
- C. Ground all raceway systems and equipment enclosures. Where not otherwise indicated, grounding conductor size shall conform to the most stringent of the governing codes.
- D. Ground the system to exterior ground rods and building cold water piping if any and ground ring shown. Provide at each such grounding location two 5/8-inch diameter by 10-foot long copper-clad steel ground rods spaced 6 feet apart and connected to the system neutral.
- E. Furnish an additional 30 feet of #2 AWG CU ground ring as a supplemental electrode connected to panel ground.
- F. Furnish bolted grounding conductor connections. All grounding points to be accessible after installation. A ground rod enclosure shall be provided flush at final grade with a lid to access ground connection.
- G. Furnish an insulated green copper equipment ground conductor installed in each conduit sized for the largest branch circuit per NEC 250-122, unless otherwise noted.
- H. Maintain continuity of equipment ground throughout the entire raceway, cabinet and equipment enclosure system. Use grounding bushings and jumpers wherever normal conduit termination does not ensure continuity. Where nonmetallic conduit is used for distribution or where direct burial cables are employed, install a green with one or more yellow stripes, insulated equipment ground conductor with each circuit.
- I. At all sheet steel enclosures such as panel boards, load centers, intermediate junction boxes provide grounding locknuts with a ground conductor termination lug on each locknut. Bond all grounding locknuts together to enclosure with copper conductors. Boxes less than 12" x 12" are not required to have conduit bonding, provide setscrew bonding locknuts. Outlets boxes 4"x4" and smaller may have locknuts only.
- J. Bond metal parts of lighting fixtures not otherwise grounded by bolted fastenings to conduit system with bare copper wire.
- K. Ground receptacles to outlet boxes with wire and machine screw.

3.15 PULL VAULT GROUNDING

- A. Provide each electrical buried pull vault with two copper-clad steel ground rod, 5/8-inch diameter by 10 feet long, driven through the floor of the splice vault leaving 6 inches projecting above the vault floor.
- B. Where pull vaults are provided provide ground rods in each vault. Details on the drawings are to establish the general scope of work.
- C. Provide two 10 ft ground rods in each primary power pull vault in drain holes.

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3.16 PRIMARY CONDUCTOR INSTALLATION

- A. Coordinate with the Utility Provider Intercounty Electric for installation of primary conductors.

3.17 TESTS

- A. General: Perform the tests as described below. Notify the COR in writing five days in advance of all tests in this and other sections of this specification. Perform all tests in the presence of the COR. Upon completion of all tests, submit written test results in duplicate for approval by the COR prior to acceptance.
- B. Test Report Forms:
1. Submit a copy for approval of the proposed tabular data sheet to be used for each required test. Type column and heading titles on sheets. Submit data sheets for approval a minimum of 10 days before each test. Divide sheets into columns and rows with appropriate rulings and titles. Enter test data on tabular forms during tests by hand; do not recopy.
 2. Furnish data in tabular form listing required test data, name of tested device, location, serial number, type and name of measurement instrument with scale settings.
- C. Preliminary Tests (distribution panels and motor control panels)
1. Install all mechanical and electrical equipment before any preliminary electrical tests are made and final connections are made.
 2. Test all control and power conductors including neutrals for insulation resistance and continuity as described below and in Section 260400 (before connection of conductors to control panels, controlled and pilot devices).
 3. Connect all power to distribution panels after all preliminary tests have been done as described above.
- D. Ground System Tests: Perform ground-impedance measurements utilizing the fall-of-potential method in accordance with IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test. Provide site diagram indicating location of test probes with associated distances and provide a plot of resistance vs. distance. Measured resistance from rod to earth shall be 25 ohms or less and less than 15 ohms for a ground ring measure at point of connection.
- E. Insulation Resistance: Test insulation resistance under normal climatic conditions. Conform to the following:
1. Measure resistance between all insulated conductors (including neutral and isolated ground) and ground as well as each other. Coordinate testing with COR so that tests can be witnessed if desired.
 2. For circuits with 600 V rated insulation, make insulation resistance measurements on each power distribution and control panel, main feeder and each branch circuit and all control circuits. Make all measurements with loads disconnected and after all wiring is in place. Perform tests with Megohm meter having a minimum 10,000 megohm scale range and apply 1000 VDC. Minimum resistance shall be 200 megohms.

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- F. Insulation Resistance Testing Procedure 600V rated wiring
1. Object of test: Measure the resistance of all insulated conductors to ground and to each other. This includes each power distribution and control panel, main feeder and each branch circuit and all control circuits.
 2. Perform testing after all wiring is in place, with loads disconnected. All electronic devices must be disconnected to avoid damage. This includes electronic switches and outlets (GFI's). GFI breakers in branch panels must be removed, with the phase and neutral wires connected to an adjacent breaker for testing.
 3. Open the main service switchgear disconnect switch.
 4. Close the branch panel main switch and all branch circuit breakers.
 5. Connect Megohm meter to phase, neutral, ground and isolated ground busses in the combinations shown in the following table. The neutral and isolated ground must be disconnected from ground at the main service switchgear in order to test the feeders.
 6. The following table can be used to record readings:

TEST LOCATION:						
	A	B	C	N	G	IG
A	n/a					
B	n/a	n/a				
C	n/a	n/a	n/a			
N	n/a	n/a	n/a	n/a		
G	n/a	n/a	n/a	n/a	n/a	
(Minimum of 10 readings w/o isolated ground)						
(Minimum of 15 readings with isolated ground)						

7. Apply voltage to circuit:
 - a. Wires with 600 V rated insulation: Apply 1000 VDC. A minimum value of 200 megohm is required.
 - 1) Exception. Outlets may arc at less than 1000 VDC. If this happens, reduce the voltage to a lesser value and record the megohm value.
 - b. Wires with less than 600 V rated insulation: Apply voltage which does not exceed manufacturer's recommendations. A minimum value of 200 megohm is required.
 8. If the megohm value is less than 200 megohms, isolate the problem circuit by opening breakers one at a time.
- G. Test Failure: Failure of any phase of the operation tests shall constitute failure of the system. Correct failures and retest system. Repeat until the system is operating to required specifications.
- H. Test Reports
1. Submit dated "Electrical System Test Reports" indicating all tests performed and demonstrating conformance with the required system performance criteria in tabular form as described above. Include all voltage, current, and resistance test data of the electrical service, main feeders, panelboards, motor control panels, control system tests and ground systems as applicable. Submit test reports prior to final inspection.
- I. Final Corrections:
1. Correct promptly any failure or defects revealed by these tests and re-conduct tests on these corrected items. Make any final adjustments for optimum operation of all equipment.

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END OF SECTION 260010

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SECTION 260050 – BASIC MATERIALS AND METHODS

PART 1 - GENERAL

1.1 REFERENCE DOCUMENTS

- A. 013300 – Submittal procedures
- B. 260010 - General Technical Provisions
- C. 260400 – Underground Power Distribution

1.2 DESCRIPTION

- A. Furnish all new materials of latest standard design, which are the standard product of manufacturers who regularly engage in the production of such equipment. Provide material of the same type or classification and used for the same purpose of the same manufacturer. Furnish materials, equipment, and devices approved by the Underwriters Laboratories, Inc., for the purposes for which they are used. Certain classes of materials used are not within the scope of UL Reexamination Services; therefore, this does not apply when not applicable.

1.3 SUBMITTALS

- A. Submit under provisions of Section 013300-Submittals.
- B. Submit shop drawings and equipment catalog data and manufacturer's installation operating and maintenance instructions for approval for the following items in accordance with Section 260010 and Division 1 submittal requirements.
- C. Provide manufacturer's original PDF files with submittals in electronic form with items marked to be submitted with all accessories. Items to be submitted include but not limited to;
 - 1. Electrical System, Panel boards and Circuit Breakers – Shop Drawings. Shop drawings shall include, an overall system drawing, one-line fault current diagram with system fault calculations and Short Circuit Current Requirements (SCCR) determine and shown for all panelboards, disconnects and other equipment where a SCCR is required. Calculate and show Ampere Interrupt Current (AIC) for main service breakers and breakers protecting feeders and motor branch circuits. Calculations shall be performed after coordination with the electric utility provider to determine the supply side transformer sizes and impedance. For bidding purchases the contractor shall base cost on requirements shown in the drawings. If there are significant differences the Contractor may request an equitable adjustment from the Contracting Officer.

Panelboard drawings shall provide detail of the size, materials, and ratings of all panel board construction components including circuit breakers. A listing of compliance standards shall be provided.

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Shop drawings shall be prepared under the direction of a Missouri licensed Electrical Engineer and include the Engineers stamp and signature.

2. Panel Boards and (over Current Protection Devices- Circuit Breakers - Product data, specification sheet, listing of compliance standards, installation instructions and Operations and Maintenance Manuals.
3. Conduit and Raceways – Product data, specification sheet, compliance certificates.
4. Pull and Junction Boxes - Product data, specification sheet, compliance certificates.
5. Conduit Fittings - Product data, specification sheet, compliance certificates.
6. Switches, Disconnect, Receptacles and Plugs - Product data, specification sheet, compliance certificates.
7. Conductors - Product data, specification sheet, compliance certificates and 1-foot samples.
8. Identification Nameplate Schedule – Product Data
9. Wire Number Tags – Product data
10. Lugs and Connectors- Product data, specification sheet, compliance certificates.
11. Insulated Terminal Ends - Product data, specification sheet, compliance certificates.
12. Ground rods - Product data, specification sheet, compliance certificates.
13. Ground rod enclosures - Product data.
14. Duct Seal and Fire Stopping - Product data, specification sheet, compliance certificates.
15. Oxide inhibiting compound - Product data, specification sheet.

PART 2 - PRODUCTS

2.1 CONDUIT AND RACEWAY

- A. Intermediate Metal Conduit (ANSI C80.3)
- B. Electrical Metallic Tubing ANSI C80.3.
- C. Rigid PVC UL listed Schedule 40 or Schedule 80
- D. Galvanized Flexible Metallic Conduit ANSI/NFPA 70.
- E. Flexible Metallic Conduit ANSI/NFPA 70.
- F. Liquidtight Flexible Metallic Conduit ANSI/NFPA 70 additionally provide liquidtight flexible conduit, zinc-coated steel core, extruded gray PVC cover, U.L. approved.

2.2 PULL AND JUNCTION BOX

- A. Provide pull boxes at every 500 feet along buried conduit corridor.
- B. Provide code gage galvanized sheet steel NEMA 1, 3, 4, 4X SS, 7, 12 pull boxes where required or shown on the drawings. Provide removable screw cover on the largest access side of the box unless otherwise detailed. Where cast boxes are required or specified, provide conduit entrances with threaded hubs. Provide stainless steel screws at all exterior and damp locations. Where pull boxes are required but not shown on the drawings, provide pull boxes as specified above sizes per NEC requirements or at Contractor's option, utilize OZ Type PB, PBW, or PFV pull box

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fittings. Paint all pull boxes with a rust inhibiting primer and finish coat to match surrounding finish.

2.3 SPLICE BOX

- A. Provide Quazite Underground Enclosure Assembly-PG1118Z80217 or approved equal. Cover logo to say "Electric".

2.4 COVER PLATES AND ENGRAVING

- A. Provide impact resistant white, nylon cover plates for all wiring devices in finished space unless otherwise indicated.
- B. Receptacle nameplates shall have panel and circuit number engraved at top of plate in 0.25 inch high letters. Engraving shall be filled with black enamel for ivory plates, white enamel for brown plates and ivory plates filled with orange paint for isolated ground receptacle.
- C. Cover Plates F.S. W-P-455. Provide anodized aluminum plates for flush mounted devices in unfinished spaces. Where devices are installed in exposed fittings or boxes, use Appleton "FSK" covers. Where weatherproof devices are specified in exterior or damp location use cast malleable covers with gasket and stainless-steel screws.

2.5 CONDUCTORS

- A. This specification covers all conductors not specifically specified in other sections. Furnish conductors and cable conforming to UL, Federal Specification J-C-30, or IPCEA as applicable. Provide new cable manufactured within 1 year prior to installation. Deliver to jobsite in original cartons bearing U.L. label.
- B. 600V Power, Lighting and Control Circuits: Provide copper conductors. Aluminum conductors may be used for underground feeders and service conductors where shown on the drawings. See Section 26010 for conductor requirements. Copper conductors unless otherwise specified, conform to Federal Specification J-C-30. For cable type THWN, provide insulation conforming to Federal Specification J-C-30A. For type XHHW, XLP-USE-2, provide cross-linked polyethylene insulation conforming to IPCEA S-66-524 9See 260400 for additional details and testing). Color code insulation for wire sizes No. 14 through No. 6, No. 4 AWG and larger may be color coded with tape. See 26010 for specific conductor types.

2.6 IDENTIFICATION NAMEPLATES

- A. Provide permanently mounted laminated plastic engraved nameplates for all electrical equipment listed in 260010 and this section indicating name, function and rating. Submit nameplate legends for approval with shop drawings of equipment. Furnish black nameplates with engraved white letters for 120/240, 120/208-volt, and red with white letters for 480-volt equipment. Letter height 1/4 inch minimum.
- B. See 260010 for required nameplates and mounting methods.

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2.7 WIRE NUMBERS

- A. Provide vinyl cloth, machine made numbered adhesive labels for each termination of all power and control conductors. Numbered plastic slip-on sleeves, or crimp-on sleeves are also acceptable. Number labels to correspond to schematic and wiring diagrams for each conductor. Provide labels with permanent adhesion method. Hand lettered labels are not acceptable.

2.8 LUGS, CONNECTORS AND TERMINATIONS

- A. F.S. W-S-610B
- B. Solderless insulated pressure twist-on connectors for conductor's No. 8 or smaller for in-line splices for circuits other than motor power and control circuits.
- C. Use circumferential compression fittings for No. 6 AWG and larger conductors, termination lugs and splices. Copper conductors - tin plated copper terminals filled with oxide inhibiting pentrox compound.
- D. Compression fittings shall be applied with a factory matched die in a hydraulic compression tool that stamps manufacturers name when fully compressed to manufactures specifications.
- E. All feeder terminations shall be rated at 75 degrees C. or higher.

2.9 INSULATION

- A. Provide insulation equal to insulation of conductors used. For above ground installations, use thermoplastic tape, heat shrink tubing, pre-insulated splices, expandable insulated sleeves or resin encapsulated splices.

2.10 TAPE

- A. Tape F.S. HH-I-595B.

2.11 METER BASE

- A. Per requirements of the public utility provider.

2.12 MOUNTING HARDWARE

- A. NSC threads on all threaded fasteners, nuts, bolts, screws, lag bolts, expansion shield, and other miscellaneous fasteners. Provide cadmium plated steel hardware for indoor dry locations.
- B. Provide all stainless-steel hardware, nuts bolts washers etc. for all outside locations and all wet or corrosive locations.

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2.13 INTERIOR PERFORATED MOUNTING CHANNEL

- A. Provide 1-1/2" x 1-1/2" continuous steel channel with 17/32" diameter holes on 1-1/2" centers on 3 sides of channel for bolting to. Fourth side of channel shall have 7/8" continuous slot for accessory hangers. Furnish accessory mounting hardware for equipment, clamps, hangers, etc.

2.14 EXTERIOR PERFORATED MOUNTING CHANNEL

- A. Provide 1-1/2" x 1-1/2" continuous stainless-steel channel as above. Provide stainless steel clamps, washers and screws.

END OF SPECIFICATION 260050

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SECTION 260400 – UNDERGROUND POWER DISTRIBUTION

PART 1 - GENERAL

1.1 DESCRIPTION

- A. These specifications provide for installation of underground secondary distribution, underground power conductors using trenching methods.

1.2 REFERENCES

- A. 260010 General Technical Provisions
- B. 260050 Basic Materials and Methods

1.3 SUBMITTALS

- A. Submit shop drawings and equipment catalog data and manufacturer's installation operating and maintenance instructions for approval of the following items in accordance with Section 13300 Submittal procedures.
- B. Submit copies of the following:
 - 1. Conductor and Cable: Catalog data, cuts showing conductor and cable construction, thicknesses and material. Provide one-foot sample of each cable or conductor with submittals provide sample that shows manufacturer, catalog number size and other ratings.
 - 2. Power Splices: Catalog data showing construction of all materials and installation procedures.
 - 3. Splice Vault: Catalog data and shop drawings.
 - 4. Transformer, Switchgear: Complete electrical drawings, shop drawings, electrical parameters including percent resistance, percent reactance and percent impedance, overload rating, voltage, and taps. Transformers tests in accordance with American Standard Test Code for Transformers.
 - 5. Metallic Core Marking Tape: Catalog Data.
 - 6. Tracer conductor: Catalog data
 - 7. Nameplates for cable identification provide engraving list with submittal.

1.4 QUALITY ASSURANCE

- A. Follow the latest edition of the National Electrical Safety Code and NEC except where local regulations are more stringent.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Protect all material and equipment to be incorporated in construction from the elements.

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- B. Handle cable carefully at all times to avoid damage. Excessive bending of the cable will not be permitted.
- C. Always seal the ends of the cable against moisture with suitable end caps. Where it is necessary to cut the cable, terminate or seal ends immediately after the cutting operation.

PART 2 - PRODUCTS

2.1 CONDUITS

- A. Install all power in conduit as shown on drawings. Where conduit size is not indicated size based on NEC requirements.
- B. Non-metallic conduit and fitting shall be electrical grade, UL Listed, schedule 40 PVC and shall conform to applicable sections of NEMA TC2-latest edition and UL 651. Any PVC conduit not having the proper NEMA and UL markings will not be accepted. Only gray-colored conduit will be accepted.
- C. Coilable Duct for directional boring installations shall be high density polyethylene (HDPE) in 3", 4", or 5" nominal inside diameter, as specified on the Drawings and shall meet ASTM D-1248. The coilable duct shall be black with three UV-resistant red stripes extruded into the outside surface. The stripes shall have a thickness in range of 10-15 mils and be 3/16" to 3/8" wide. The stripes shall be evenly spaced at 120° intervals around the circumference of the duct.

2.2 POWER CONDUCTORS – UNDERGROUND INSTALLED CONDUIT

- A. Conform to REA Specification (XLP-USE-2 insulation, cross-linked polyethylene) U-2 Type 2 for 600 volts secondary unless otherwise shown on the drawings, and as described below.
- B. All conductors shall be copper or aluminum as indicated on the drawings.

2.3 SECONDARY POWER CONDUCTORS (600V AND LESS)

- A. Meets REA and IPCEA S-95-658 & Nema WC70 with additional features as described below:
 - 1. Listed for RHH, RHW-2, USE-2
 - 2. Standards: UL 854, ASTMB3, ASTMB8 Federal Spec A-A-59544
 - 3. Conductor: Copper annealed coated, Class B Stranded.
 - 4. Conductor: Aluminum complying with ASTM B 800 and ASTM B 801
 - 5. Insulation: XLP-USE chemically cross-linked thermosetting polyethylene.

2.4 UNDERGROUND POWER CONDUCTOR CONNECTORS AND SPLICES (600V)

- A. Utilize 3M Company "Scotchcast" urethane resin insulation in poured kits, splices and compression connectors for underground direct burial cable No. 12 to 750 MCM or approved equal.

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- B. Provide Scotch 3M 82A series poured resin splice enclosures or approved equal for straight inline splices
- C. Provide Scotch 3M 90-B1 enclosures for “Y” tap splices
- D. Provide Scotch 3m Multimold 85-CP kit for larger splice enclosures.
- E. Provide molded plastic enclosure, pouring funnels, closure tape for splices kits.
- F. Provide 3M™ Scotchcast™ Electrical Insulating Resin 4, two-part epoxy insulating, and encapsulating resin used to splice solid cables up to 8 kV.
- G. Provide compression lugs for all splices.
 - 1. Provide compression lugs with hydraulic compression tool die which imprints manufactures name when splice is properly compressed. Fill all lugs with Penetrox before crimping splice.

2.5 SPLICE VAULT / PULL VAULT (POWER AND CONTROL CABLE)

- A. Provide precast concrete splice vault approximately 4' x 4' x 4', 2905 lbs. with hinged and spring locked steel cover, 36" x 36" Oldcastle 444-LA with 44-332-P cover or approved equal.
- B. Provide precast concrete splice vault approximately 2' x 2' x 2', weight approximately 800 lbs. with hinged and spring locked steel cover, 24" x 24" Oldcastle 3030-LA with 44-332-P cover or approved equal. Provide locking latch mechanism to accept padlock. Provide traffic rated lid or approved equal.

2.6 METALLIC CORE MARKING TAPE

- A. Furnish plastic covered aluminum marking tape for power and control conductors.
- B. Tape width: 2".
- C. UNDERGROUND DETECTABLE - 5 MIL 2 Mil Polyethylene Film Reverse Printed And Laminated To Aluminum Foil Then Laminated To 2 Mil Polypropylene Film
- D. Color: Power- Red, Control Cable-Orange.
- E. Inscription: Power Conductors "Caution, Buried Power Line".
 - 1. Red- Buried Electrical Line Below

2.7 TRACER CONDUCTORS

- A. Provide copper insulated tracer conductors for marking location of underground conduits.
- B. Conductor shall be #10 AWG, XLP-USE insulation 0.045” thick.
- C. Conductors shall be installed above conduits and tie wrapped to conduit.

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PART 3 - EXECUTION

3.1 INSPECTION AND INVENTORY OF BURIED POWER AND CONTROL CABLE

- A. Before any backfilling operations are begun, the Contractor and COR shall jointly inspect all trenches, cable placement, splice vaults, risers, marking tape, tracer conductors and other construction not accessible after backfilling.

3.2 PREPARATION

- A. Trenching and backfill for electrical conduits shall conform to details shown in the drawings.

3.3 POWER AND CONTROL CABLE INSTALLATION

- A. Directionally boring:
 - 1. Coilable duct shall be directionally bored at standard burial depths of 36" of cover for ducts housing primary cables and 24" to 36" of cover for ducts housing secondary cables.
 - 2. Duct Joining: Sections of coilable duct shall be joined using fusion couplings. Where the coilable duct is to be joined with PVC conduit, the Contractor is to fuse a 24" straight piece of matching size PVC conduit to the coilable duct. Prior to fusing, the Contractor shall make circumferential scores around the outside of the PVC conduit at the end that is to be installed into the fusion coupling. This will allow the fusion coupling to adhere to the dissimilar PVC conduit.
 - 3. Vertical and Horizontal Bends: All vertical bends shall be made using Schedule 40 PVC bends with belled or coupled ends. Under no circumstances shall a vertical bend be fabricated by "sweeping" it up to grade using the guided boring machine. The vertical bend shall be joined directly to a 36" straight piece of matching size PVC conduit, which is installed to transition between the bend and the coilable duct. Horizontal bends, however may be swept using the guided boring machine. In places where the radius is too small for this, a PVC bend shall be installed between two 36" straight PVC transition sections.
 - 4. Connections into Vaults and Manholes: Bored coilable duct shall end 4'-5' from a new vault or manhole. This coilable duct shall then be transitioned to the same size PVC before connecting into the structure. The transition to PVC shall be made using the techniques described above in 3.3.B.2 – Duct Joining.
- B. Conduit Cleaning
 - 1. Prior to pulling cable thru conduits clean all conduits by swabbing out conduits to remove all debris. Cap ends after cleaning prior to installing conductors.
- C. Placing of Cable in Trenches: Do not lay cable when the outside temperature is below the minimum temperatures specified by the cable manufacturer.
 - 1. Power, communications cable and fiber optic cable shall be installed in conduit only. Direct burial is not allowed.
 - 2. Wherever possible, cable shall be played out from the reel mounted on a moving vehicle or trailer. Support the reel on a stand. Place cable in the trench by hand.
 - 3. Inspect the cable as it is removed from the reel in laying operations to be certain that it is free from visible defects.

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4. Cable must be pulled through conduit; duct or other locations shall be lubricated with a suitable cable lubricant approved by cable manufacturer prior to pulling into conduit or duct.
 5. Do not exceed cable pulling tension as recommended by manufacturer.
 6. Provide a pulling dynamometer to measure pulling tension on all pulls.
 7. Leave a minimum of 24" slack on each cable at all risers, splices, transformer pads, pedestals and terminal points.
- D. Cable Minimum Bending Radius: Measure radius to the surface of the cable on the inside of the bend.
1. Secondary service cable (600V or less) 6 times the overall diameter of the cable.
 2. Primary cable (more than 600V) 12 times the overall diameter of the cable.
 3. Control cable - two feet radius.
- E. Splices: Not more than one splice for each 2000' of power or control cable installed in any one continuous run.
1. Make Cable splices in accordance with the splice manufacturer's instructions.
 2. Use pre-formed molded case poured "potted" splice kits installed in accordance with manufacturer's recommendations.
 3. Make all splices in splice vaults or at pedestals.
 4. Bends will not be permitted within 12 inches of the ends of splice.
 5. The cable or circuit numbers and the exact location of the splices shall be noted and given to the COR.

3.4 PULLING STRESSES

- A. Maximum 500 pounds as measured by a dynamometer or less if recommended by cable manufacturer.
- B. Do not use pulling eyes or other attachments connected directly to the conductor.

3.5 MARKING TAPE INSTALLATION

- A. Install metallic core marking tape above each electrical conduit as shown on drawings

3.6 TRACER CONDUCTOR INSTALLATION

- A. Install tracer conductor above each electrical conduit.
- B. Tracer conductor shall be tie wrapped with nylon straps to conduit every 10'.
- C. Bring conductor up above grade at each termination and in each pull or splice vault. Coil 6' of conductor at each location.

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3.7 BACKFILLING

- A. Conform to details shown in the drawings or as required by the Electric utility provider for primary conductors.

3.8 CONDUIT

- A. All exposed ends of conduit shall be plugged during construction.
- B. Remove burrs or sharp projections that might injure the cable.

3.9 TERMINATIONS OF POWER CABLE

- A. Make terminations in accordance with the manufacturer's instructions.

3.10 BENDS

- A. No bends shall be permitted within six inches of the ends of stress cones.

3.11 GROUNDING

- A. Grounding shall conform to Section 260010.
- B. Provide a suitable ground installation at each panel board, panels, subpanels, pedestal locations and all other locations required by the NEC.
- C. Solidly connect the lead from the grounding electrode to the primary neutral, unless shown otherwise on the drawings, the secondary neutral and the ground connection of all equipment at the location.
- D. For ground test see Section 260010.

3.12 IDENTIFICATION OF CABLES

- A. In each pull / splice vault and at each termination, provide engraved plastic nameplates tie wrapped to conduit in all splice and pull vaults or to wall if in buildings to identify the cable as follows on each nameplate:
 - 1. Circuit Voltage, phase, cable size either in AWG or control cable pair number and AWG size, source and termination of each cable.
- B. See 260010 for nameplate construction and lettering height

3.13 CABLE MARKERS

- A. Mark power and control cable location with permanent cable markers 200 feet maximum interval and at all changes in direction greater than 30 degrees.

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- B. Provide cable direction marker.
- C. Install two-inch wide labeled metallic core plastic warning tape above cables as shown on drawings.
- D. Provide locating conductor above each conduit as described above and below

3.14 POWER CABLE TESTS

- A. Conduct tests in the presence of the COR on each length of underground power in the system, with terminations in place but disconnected. See Section 260010 TESTS for test report format and general provisions. These tests apply to cable direct buried or pulled in conduit.
- B. Provide an independent testing firm(s) that specializes in testing power and telephone cable, with demonstrated minimum of 3 years in testing cable. Provide the appropriate testing equipment for high-pot, continuity and insulation resistance testing, and for fault locating.
- C. Preliminary Continuity and Insulation Resistance (Power Cable): After installation of the power cable and prior to backfilling and the high potential test specified below, perform a simple continuity test on the system by shorting each conductor pair together at the source and checking for continuity from the end of each tap or line with an ohmmeter. Megger each conductor to ground (includes neutral) and to each other conductor to test for insulation faults prior to doing hipot tests below. See 260010 PRELIMINARY TESTS for insulation requirements. Ground one lead to a 5-foot driven ground rod and connect the other lead to conductor with terminations disconnected. Measure insulation resistance to ground and between phases with a DC Megger, 2,500 VDC, prior to doing hi pot test described below. After backfilling perform high - pot tests described below.

3.15 HIGH POTENTIAL (POWER CABLE)

- A. After successful continuity, insulation resistance tests and backfilling, perform high potential tests on each length of cable in the system.
- B. Maintain for a minimum of five minutes a DC test potential voltage as follows:

- C. 600-volt secondary cable
 - 1. XLP-USE Insulation

Size	Insulation Thickness In.	Field DC Acceptance Voltage
AWG or MCM	Inches	Test Volts
12	.0450	4.8KV
10	.0450	4.8 KV
8-2	0.060	4.8kV
1-4/0	0.080	6.4 KV
250-500 MCM	0.095	8.0 kV
600-1000 MCM	0.110	9.6 KV

- D. The voltage may either be increased continuously or in steps to the maximum test value.

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- E. If increased continuously, increase uniformly to maximum voltage in not less than 10 seconds and in not more than approximately 60 seconds. Apply maximum test voltage for a total of five minutes.
- F. If applied in steps, increase from one step to the next uniformly with each step long enough for the absorption current to attain reasonable stabilization (one minute minimum). Take current and voltage readings at the end of each step duration. The number of steps shall be from five to eight.
- G. Plot leakage current in microamps DC vs. test voltage. Continue plot for five minutes.
- H. Correct point faults by enclosing fault area in cast in-line splice enclosure, WITHOUT SPLICING. If wire is damaged remove and replace failed conductor with in-line compression splice lugs and poured urethane splice as above.
- I. Install any splice or fault area described above in a splice vault or pedestal.
- J. Submit all test results in hard copy and PDF.
- K. All faults discovered by this test shall be corrected and the tests rerun until cable splice meet requirements.
- L. Warning: Following these tests, a hazardous voltage may exist on the power cable. The conductor shall be grounded to permit any charge to drain to earth.

END OF SECTION 260400

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SECTION 329200 – SEEDING AND MULCHING

PART 1 - GENERAL

1.1 SUMMARY

- A. This section includes site rehabilitation work, including seeding and mulching.

1.2 SUBMITTALS

- A. Product Data: For each type of product indicated
- B. Product certificates.

1.3 DELIVERY, STORAGE, AND HANDLING

- A. Seed: Deliver, store and handle materials according to manufacturer's recommendations. Materials must be kept dry until ready to use.

PART 2 - PRODUCTS

2.1 SEED

- A. Grass Seed: Fresh, clean, dry, new-crop seed complying with ASOA's "Journal of Seed Technology; Rules for Testing Seeds" for purity and germination tolerances.
- B. Grass Seed Mix – the following seed mixes will be used to stabilize the soil and revegetate the site:
 - 1. If applied during November through March:

Sites will be revegetated using a seed mix containing River Oats (*Chasmanthium latifolium*), Little Bluestem (*Schizachyrium scoparium*), Showy Tickseed (*Bidens aristosa*), Partridge Pea (*Chamaecrista fasciculata*), Lance-Leafed Coreopsis (*Coreopsis lanceolata*), Wild Bergamot (*Monarda fistulosa*), Slender Mountain Mint (*Pycnanthemum tenuifolium*), and Black-Eyed Susan (*Rudbeckia hirta*).

 - Seeds shall be from Missouri eco-types or, if unavailable, from adjacent states. Seeding shall be at a rate recommended by the vendor.
 - In areas where slope is greater than 35%, then an annual rye and/or winter wheat and weed free straw shall be applied at 50 pounds per acre.
 - 2. If applied during April through October:

Sites will be revegetated using an annual rye and/or winter wheat mixed with Little Bluestem (*Schizachyrium scoparium*) applied at 50 pounds per acre.

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2.2 PLANTING MATERIALS

- A. Topsoil Source: Reuse surface soil on-site and amended topsoil from commercial sources. Provide topsoil which is fertile, friable, natural loam, surface soil which is reasonably free of subsoil, high clay content, brush, weeds, litter, stumps, rocks and any other material which may be harmful to plant growth. Topsoil shall have a pH range of 5.5 to 7.4.

NOTE: The C.O.R. shall approve the Contractor's provided soil sample prior to installation and planting

- B. Mulches:
1. Straw Mulch: Provide air dry, clean, mildew- and seed-free, and certified weed-free threshed straw of wheat, rye, oats, or barley.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Finish Grading: Grade planting areas to a smooth, uniform surface plane with loose, uniformly fine texture. Grade to within plus or minus ½ inch of finish elevation. Roll and rake, remove ridges, and fill depressions to meet finish grades. Limit fine grading to areas that can be planted in the immediate future.
- B. Restore areas if eroded or otherwise disturbed after finish grading and before planting.

3.2 SEEDING

- A. Scarify existing subsoil and compacted areas prior to spreading topsoil. Prepare subsoil to eliminate uneven areas. Maintain profiles and contours. Make changes in grade gradual. Blend slopes into level areas.
- B. Spread organic topsoil at a minimum depth of 4 inches over area to be planted and integrate with the top layer of subsoil to prevent hardpan separation. Rake the surface smooth.
- C. Place topsoil during dry weather and on dry, unfrozen subgrade.
- D. Remove foreign, non-organic material from topsoil while spreading.
- E. Grade topsoil to eliminate rough, low or soft areas and to ensure positive drainage.
- F. All seeded areas to receive a settled minimum depth of 4" of topsoil.
- G. Remove foreign materials, weeds and undesirable plants and their roots. Remove any contaminated subsoil if present.
- H. Prepare seeding areas by removing and/or killing off any unwanted existing vegetation with a glyphosate herbicide, applied only by a state certified applicator no sooner than 2 weeks prior to seed installation. Soils surface should be loose and free of any soil clumps exceeding 1 inch in diameter.

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- I. Seed shall be installed by means of mechanical and/or hand broadcast methods to ensure even distribution of seeds throughout all designated seeded areas. Native seed mix shall be mixed thoroughly by vendor/supplier at their recommended ratios.
- J. Apply seed evenly in two intersecting directions. Rake in lightly, then lightly mulch with clean, weed free straw. Seeded areas shall be watered immediately following installation to accelerate cover crop germination.
- K. An acceptable quality seed installation is defined as having:
 - 1. No bare spots larger than 1 square foot after substantial germination.
 - 2. No more than 10% of the total area with bare spots larger than 1 square foot.
 - 3. A uniform coverage throughout all seeded areas.
- L. Do not perform seeding in windy conditions.
- M. Surface layer of soil for seeded areas shall be kept moist throughout germination period. Mulch the areas with a light covering of clean, chopped straw to retain moisture and use a tackifier to prevent wind damage. If installed in spring, lightly water 4-6 weeks after germination at regular intervals (depending on rainfall), or if an extended drought occurs throughout the summer months.
- N. Protect seeded areas from pedestrian traffic or vehicular trespassing with temporary fences, signs, barriers, etc. while seed is germinating. Repair damage resulting from trespass, erosion, settlement, etc. Remove temporary protective devices after final acceptance.
- O. Maintain seeded areas immediately after placement until vegetation is well established and exhibits a vigorous growing condition.
- P. Control the growth of weeds to the extent possible by using various methods such as line trimming, mowing and selective herbicide use. Apply herbicide in accordance with manufacturer's instructions. Contractor will be responsible for any damage resulting from improper use of herbicides.
- Q. Immediately reseed areas that show bare spots.

END OF SECTION 329200

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APPENDIX A

Well Pump Test and Recovery Data Sheet

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Well Pump Test and Recovery Data Sheet

Project name/Location of pump test: _____

GPS latitude: _____ GPS longitude: _____ GPS elevation: _____

WSSN: _____ Well classification: _____

Depth pump set: _____

Well casing height above ground: _____

Well depth to top of casing: _____

Water levels measured by: _____ Pump: _____ Electric meter: _____ Other: _____

Pump test conducted by: (person's name, not just the company): _____

Water meter reading at beginning of test: _____ gallons

Water meter reading at end of test: _____ gallons

Total gallons pumped: _____ gallons

Time starting the test: _____ Static water elevation: _____ Time test ended: _____

Date test started: _____ Date test ended: _____

Well shall be pump tested to determine its maximum pumping rate at a stabilized drawdown which does not exceed 1.5 times the design pumping rate (15 gpm) or the maximum amount the screen (if present) can safely pass.

Time Elapsed in minutes	Pumping Rate in GPM	Pumping Water Level in feet and inches measured from top of casing	Comments
0			
2			
4			
6			
8			
10			
15			
20			
25			
30			
40			
50			
60			
75			
90			
105			
120			
150			
180			
210			
240			

Well Recovery Data

Time Elapsed in minutes	Pumping Rate in GPM	Pumping Water Level in feet and inches measured from top of casing	Comments
0	Stopped		This reading starts as soon as the pumping stops.
2	Stopped		
4	Stopped		
6	Stopped		
8	Stopped		
10	Stopped		
15	Stopped		
20	Stopped		
25	Stopped		
30	Stopped		This reading should be the same as the original static water level. If not, continue time and recordings until the static water level has stabilized.

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APPENDIX B

Existing Conditions Photos

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Photo 1 – View of well and bunker/wellhouse (red box) in distance.



Photo 2 – Existing generator that previously powered well pump. No longer working, leave in place.

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Photo 3 – View of pump control box and where electrical conduit enters/exits building and control box.



Photo 4 – Close up of existing pump control box. Contractor to replace existing control box to coordinate with new well pump.

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Photo 5 – View of existing electric control panel. Located in well house bunker. Contractor to connect new electric to existing electric control panel to power new well pump.



Photo 6 – View of circuit directory on electric control panel.

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Photo 7 – View of Recreation Area sign located near entrance to the “Old Road Corridor” from Hwy DD. Arrow is pointing towards the corridor.

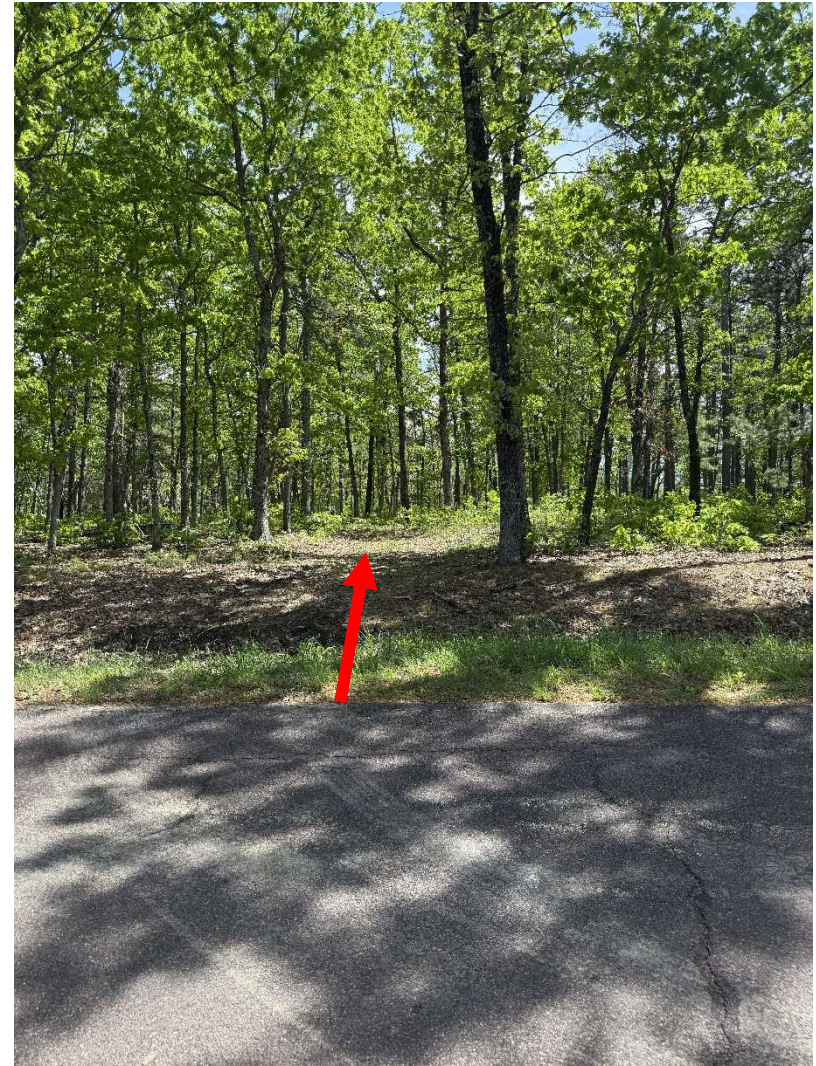


Photo 8 – View of the start of the “Old Road Corridor” from Hwy DD. Arrow is pointing towards the corridor.